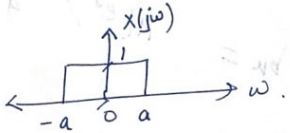
TUTORIAL - 8 Answers (EED-201)

Ans-1

(i) $x(t) = \dfrac{\sin at}{\pi t}$

$$F.T.\{x(t)\} = X(j\omega) = \begin{cases} 1, & \omega < |a| \\ 0, & \text{otherwise} \end{cases}$$

$\therefore$



(ii) $F.T.\{e^{-a|t|}\ ;\ a>0\} \longleftrightarrow \dfrac{2a}{a^2+\omega^2}$

(iii) $\dfrac{1}{a^2+t^2} \xleftrightarrow{F.T.} \dfrac{\pi}{a} e^{-a|\omega|}$

(iv) $\cos\omega_0 t \xleftrightarrow{FT} \pi\left[\delta(\omega-\omega_0) + \delta(\omega+\omega_0)\right]$

Ans-2

(i) $\text{sgn}(t) \xleftrightarrow{FT} \dfrac{2}{j\omega}$.

(ii) $e^{t}u(-t) \xleftrightarrow{FT} \dfrac{1}{1-j\omega}$

(iii) $1 \longleftrightarrow 2\pi\delta(\omega)$

(iv) $u(t) \longleftrightarrow \pi\delta(\omega) + \dfrac{1}{j\omega}$

(v) $x(t) = e^{-2|t-1|} \xleftrightarrow{FT} \dfrac{4e^{-j\omega}}{4+\omega^2}$

(vi) $\left[te^{-2t}\sin 4t\right]u(t) \xleftrightarrow{FT} \dfrac{1/2j}{(2-j4+j\omega)^2} - \dfrac{1/2j}{(2+j4-j\omega)^2}$

(vii) $t\left(\dfrac{\sin t}{\pi t}\right)^2 \xleftrightarrow{F.T} \begin{cases} j/2\pi, & -2<\omega<0 \\ -j/2\pi, & 0<\omega<2 \\ 0, & \text{otherwise}. \end{cases}$

Ans-3

$$\delta_{T_0}(t) \xleftrightarrow{FT} \omega_0 \sum_{K=-\infty}^{\infty} \delta(\omega - K\omega_0)$$

Ans-4

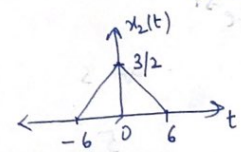
Inverse fourier transform $= x(t) = \dfrac{\sin at}{\pi t}\cos\omega_0 t$

Ans-5

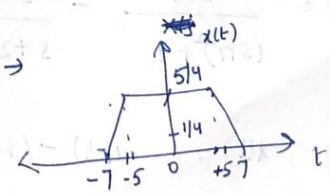
(i) $\dfrac{1}{(a+j\omega)^2} \xleftrightarrow{I.F.T} te^{-at}u(t)$

(ii) $\dfrac{\sin^2 3\omega \cos\omega}{\omega^2} \xleftrightarrow{I.F.T.} \dfrac{1}{2}x_2(t+1) + \dfrac{1}{2}x_2(t-1)$;

where $x_2(t)$ is shown below-



$\therefore$ Final ans. $\rightarrow$



Ans-6

(i) $e^{-2t}u(t) - e^{-4t}u(t) = h(t)$

$$y(t) = \frac{1}{4}e^{-2t}u(t) - \frac{1}{2}te^{-2t}u(t) + t^2e^{-2t}u(t) - \frac{1}{4}e^{-4t}u(t)$$

(ii) yes, can be solved by FT.

$h(t)$ & $y(t)$ same as (i)

Ans-7 (i) $x(t) = -e^{-at}u(-t) \xleftrightarrow{L.T.} \frac{1}{s+a}$; $\sigma < -a$

(ii) $e^{2t}u(t) + e^{-3t}u(-t) \xleftrightarrow{L.T.}$ does n't exist $\because$ ROC not common

(iii) $e^{-3t}u(t) + e^{2t}u(-t) \xleftrightarrow{L.T.} \frac{-5}{(s-2)(s+3)}$; $-3 < \sigma < 2$

(iv) $X(s) = \frac{1}{1-e^{-sT}}$; $\sigma > 0$

(v) $\delta'(t) \xleftrightarrow{LT} s \quad \forall s$

(vi) $X(s) = \frac{s+1}{(s+1)^2+4} - \frac{5}{s+2} - \frac{1/2}{s-2}$; $-1 < \sigma < 2$

Ans-8 (i) $x(t) = -u(-t) - (1+t)e^{-t}u(t)$

(ii) $x(t) = \left[\frac{-2}{13}e^{-2t} + \frac{2}{13}\cos 3t + \frac{3}{13}\sin 3t\right]u(t)$

(iii) $x(t) = -(e^{-t} + e^{-3t})u(t)$.

(iv) $x(t) = \delta(t) + (2e^{-t} + e^{-2t})u(t)$

Ans-9 (a) → (i) $H(s) = \frac{1}{(s+2)(s-1)}$

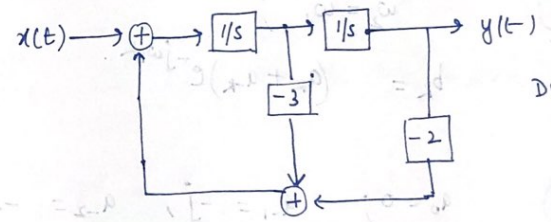
(b) (i) $h(t) = -\frac{1}{3}(e^{-2t} - e^{t})u(t)$

(ii) $h(t) = -\frac{1}{3}e^{-2t}u(t) - \frac{1}{3}e^{t}u(-t)$

(iii) $h(t) = \frac{1}{3}e^{-2t}u(-t) - \frac{1}{3}e^{t}u(-t)$

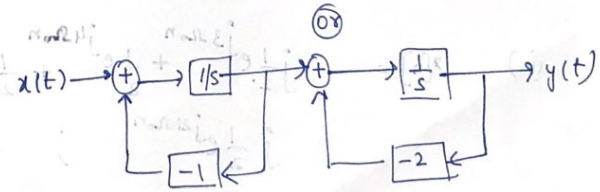
Ans-10 $\frac{d^2y(t)}{dt^2} + 10\frac{dy(t)}{dt} + 16y(t) = 12x(t) + 3\frac{dx(t)}{dt}$

Ans-11



Direct form

OR



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